Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

amsmath graphicx caption subcaption booktabs float [utf8]inputenc geometry multirow setspace parskip svg [bottom]footmisc tikz [section]placeins

trees,positioning,fit,calc block/.style = draw, fill=blue!20, rectangle, minimum height=3em, minimum width=4em, input/.style = coordinate, output/.style = coordinate

[section]minted xcolor porland

chngcntr

[hidelinks]hyperref
scrhack tocbasic lollevelup
indentfirst a4paper, margin=1in

[width=0.7]assets/birzeit-logo.svg

Department of Electrical & Computer Engineering
ENCS101 - Course Name

Report Title

Prepared by: Student's Name
Instructor: Instructor's Name
Date: 19 Pebruari 2025

Abstract

You can check the textcites.bib in order to add references (it would be automatically added to the report once you cite it[**?**] with latex[**?**]). For the figures I do not prefer static figures, I like to generate them with TikZ, but in case you've needed some static figures you can add them to the textassets directory to keep your files organized.

Daftar Isi

1 Section **1**
1.1 Tables 1
1.2 Figures 1
1.3 Equations 2
1.4 Listings 3

2 Conclusion **4**

1 Section

1.1 Tables

Tables can be built and labeled like this, adding a new row requires adding new line of entries, separated by latex and ends with latex
, while adding new column requires adding latexl in the [escapeinside=!!]latex!begintabularllll part.

Tabel 1: Table Caption

Title 1	**Title 2**	**Title 3**	**Title 4**
Entry 1	Entry 2	Entry 3	Entry 4
Entry 5	Entry 6	Entry 7	Entry 8

1.2 Figures

you can add figures with captions, also you may label them and reference them again with latex**??**.



Gambar 1.1: Birzeit Logo (PNG)

[width=0.5]assets/birzeit-logo.svg

Gambar 1.2: Birzeit Logo (SVG)

Or you can generate figures using TikZ, I've provided some styles to generate block diagrams as in Figure 1.5, but I won't be able to explain TikZ syntax for you so please try figure it out by yourself (it isn't hard to understand). Also you can add subfigures using latexsubfigure environment[1].

[b]0.45 [input] (mi) ; [block,right=of mi] (a) E $(p_k, \cdot)$; [block,right=of a] (b) D $(s_k, \cdot)$; [input,below=of a] (pk) ; [input,below=of b] (sk) ; [output, right=of b] (mo) ; [draw,->] (mi) – node[above] m (a); [draw,->] (pk) – node[left] p_k (a); [draw,->] (a) – node[above] c (b); [draw,->] (sk) – node[left] s_k (b); [draw,->] (b) – node[above] m (mo);

Gambar 1.3: Public-Key Architecture

[b]0.45 [input] (mi) ; [block,right=of mi] (a) E $(k, \cdot)$; [block,right=of a] (b) D $(k, \cdot)$; [input,below=of a] (pk) ; [input,below=of b] (sk) ; [output, right=of b] (mo) ; [draw,->] (mi) – node[above] m (a); [draw,->] (pk) – node[left] k (a); [draw,->] (a) – node[above] c (b); [draw,->] (sk) – node[left] k (b); [draw,->] (b) – node[above] m (mo);

Gambar 1.4: Symmetric-Key Architecture

Gambar 1.5: Public-Key vs. Symmetric-Key: Architecture

1.3 Equations

you can use latexalign environment to insert aligned equations like this $\tau = 28\ s$
$\omega = 2\pi\frac{2}{60} = 0.209\ rad/s$
$[t]lag = \frac{\theta(\omega)}{\omega} = \frac{\arctan(\omega\tau)}{\omega}$
$= \frac{\arctan(0.209 \cdot 28)}{0.209}$
$= 6.69\ s$

or you can insert latex at the end of each line to prevent it from being enumerated.

$\tau = 28\ s$
$\omega = 2\pi\frac{2}{60} = 0.209\ rad/s$
$[t]lag = \frac{\theta(\omega)}{\omega} = \frac{\arctan(\omega\tau)}{\omega}$
$= \frac{\arctan(0.209 \cdot 28)}{0.209}$
$= 6.69\ s$

And you can insert one line equation with latexequation environment.

$$A = 15 \cdot B \tag{1}$$

[1]Environment means a block surrounded by [escapeinside=!!]latex!beginenvname and [escapeinside=!!]latex!endenvname

and latexequation* environment to prevent the equation from being enumerated.

$A = 15 \cdot B$

And inline math if you want with latex$Expression$ like this $A = 15$.

1.4 Listings

Listing 1.4 shows a sample listing for C++ code. [H] [frame=single,framesep=10pt,breaklines,breakanywhere]C++ include <iostream> include <algorithm>

using namespace std;

int main() int n; cin » n; int a[n]; for (int i = 0; i < n; ++i) cin » a[i]; sort(a, a + n); for (int i = 0; i < n; ++i) cout « a[i]; return 0; Sample Code

Listing 1.4 shows a sample listing for shell session.

[H] [frame=single,framesep=10pt,breaklines,breakanywhere]shell $g++main.cpp$

$./a.out$432411234

Sample Output

2 Conclusion

Feel free to use this template for your reports.

Daftar Gambar

1.1 Birzeit Logo (PNG) 1

1.2 Birzeit Logo (SVG) 2

1.3 Public-Key Architecture 2

1.4 Symmetric-Key Architecture 2

1.5 Public-Key vs. Symmetric-Key: Architecture 2

Daftar Tabel

1 Table Caption 1